Modern Alchemy

a reference apparatus for Natural Knowledge

Expanded edition, with a primer
Book 3 of *Natural Knowledge - a collection*

Freddi Van Gogh

Modern Alchemy: a reference apparatus for Natural Knowledge

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How this book was made

The ideas, the questions and the decisions in this book are the author's. The author wrote the notes, the riffs and the rough drafts, and set out a recipe for each part. AI assistants turned that material into prose, checked it against sources and against the series' fact checker, and raised questions where the notes were wrong or unclear. The author answered them, made the calls, and edited the result.

In the terms publishers now use, the text is AI-generated: an AI drafted it, working from the author's own notes, research and writing as its source material. The cover was also made with an AI assistant, drawn in code.

A visual companion

Want to see these concepts in games and fun video clips? A collection of these and some more detail on visual subjects are further explored at:

https://freddi-van-gogh.github.io/Natural_Knowledge_git/data-vis-compendium/

The code can be found more quickly in the source repository, which also holds the books as PDFs:

https://github.com/freddi-van-gogh/Natural_Knowledge_git

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Gold, gunpowder, and given names: a primer on alchemy

Much of the prose on Alchemy and its history was inspired from a conversation with Dr. Zoltan Markus many years ago at a conference I won't name. He thought a fun thesis for a writing project would be connecting alchemy to chemistry, and it was indeed very fun.

This primer is about alchemy: how it was practised, how it grew into chemistry, and how its words came to describe the remaking of a person. It comes first because the rest of this book borrows alchemy's name.

Three problems

Anyone who tried to understand matter before modern chemistry faced three problems at once. I find that they explain more about alchemy than it being a sham or esoteric science akin to astrology.

The first was that much of what explains the world had not been found yet. Carl Wilhelm Scheele and Joseph Priestley each made oxygen in the 1770s. Chemists explained burning with phlogiston, Georg Ernst Stahl's name for a supposed combustible "earth", and most went on doing so until the 1790s.

The second was vocabulary. A new thing needs a name before anyone can discuss it, and the names to hand came from the theories of the day. Priestley called his gas "dephlogisticated air". In 1777 Antoine Lavoisier, who took it for an element, renamed it *oxygène*, from the Greek for "acid-maker", because he believed that every acid contained it.

Humphry Davy later showed that hydrochloric acid contains no oxygen, but the name had stuck. However, most of the acids discussed in biochemistry do contain oxygen, partly because of oxygen's properties. So Lavoisier over-simplified rather than simply erred: his rule fails for hydrochloric acid, but fits most of the acids a biochemist meets. More than a century earlier, Jan Baptist van Helmont had needed to coin the word *gas* itself.

The third was persuasion: explaining a new way of thinking to people who had no words for it and no reason to adopt it. Amedeo Avogadro proposed in 1811 that equal volumes of gases, at the same temperature and pressure, hold equal numbers of molecules. The idea was largely ignored until Stanislao Cannizzaro argued for it at a congress in Karlsruhe in 1860, four years after Avogadro's death. Sadi Carnot's short book on the power of heat, printed at his own expense in 1824, went almost unread for ten years.

Alchemy grew up inside all three problems. It had real substances, real furnaces and real skill, but few

agreed names and no sure way to pass a new idea on. So it borrowed the ideas that everyone already accepted.

Borrowing what everyone accepted

Faced with something new, people tied it to something already accepted: the Sun as a source of energy, water as a thing that moves, the Moon as a thing with a cycle. Alchemy tied its metals to the sky, pairing gold with the Sun and silver with the Moon. Sometimes such ties were right.

The Sun really does feed plants and algae. Photosynthesis stores sunlight as chemical energy in sugars and supplies most of the biological energy for complex life on Earth: about 130 terawatts, roughly seven times humanity's power use. Jan Ingenhousz showed by experiment in 1779 that plants "purify" air only in sunlight.

A plant grown in the dark can sprout and stretch, tall and pale, on the food stored in its seed. It cannot make more, though, and it starves.

Vitamin D is a different case. In the skin, ultraviolet B light breaks open a ring in a molecule called 7-dehydrocholesterol, and body heat turns the product into vitamin D3. There the light is a trigger for one reaction, not a fuel. It supplies no calories; food or supplements can stand in for it; and in Boston the winter sun made none from November to February.

The Moon was the other great accepted rhythm. Its phases repeat every 29.53 days on average, though single months vary by up to about seven hours, and its link with the sea is real. The Moon pulls harder on the near side of the Earth than on the far side, and that difference raises the tides.

The Sun adds a tide-raising force about 46 per cent of the Moon's, so the largest tides come near new and full Moon, when the two line up. Coastlines and the shapes of sea basins decide how big a tide is, and some coasts get only one a day.

Here the folk belief beat the scientist. Galileo, in his *Dialogue* of 1632, blamed the tides on the Earth's motion. Isaac Newton explained them by the gravity of the Moon and Sun in 1687.

The trouble starts when a true link is stretched. If the Sun feeds the plants, it is tempting to think it drives everything, and alchemy's gold was the Sun's metal. But the Sun fuses hydrogen into helium, and fusion in stars builds nothing heavier than the iron group.

Heavier elements, gold among them, form when atomic nuclei capture neutrons. In 2017 a collision of two neutron stars, detected as GW170817 about 130 million light years away, confirmed that such mergers forge heavy elements. Astronomers identified strontium in its light, and one model put the gold it made at around 200 times the mass of the Earth.

Most of the universe's gold was probably forged in such collisions and similar catastrophes, not in the Sun. How much came from mergers, and how much from collapsing stars or magnetar flares, is still argued over. A correspondence is not a mechanism.

Life offers a counter-example too. In 1977 the submersible *Alvin* found whole communities around hydrothermal vents, hot springs on the deep-sea floor of the Galápagos Rift. There, bacteria and archaea draw energy from chemicals such as hydrogen sulfide, and tube worms, clams and shrimp live on them.

Even these communities are not wholly free of the Sun: many vent microbes use oxygen dissolved in seawater, most of it made by photosynthesis.

A comment from the author, not a finding: the danger is in where the line gets cut. "This is because of the Sun", or of the seasons, or of the planets, or of some cultural event, is often true. It is not always the only engine behind everyday experience, or even the main one, and stopping at the first true cause hides the others.

In a way, our environment is largely supplied energy by the Sun feeding algae and plants, which are eaten by herbivores, which are eaten by carnivores, which are eaten by even larger carnivores. The problem is, it's just one of the big drivers of energy today — "solar power" being a more deliberate form of that.

What alchemists did, and how it became chemistry

Alchemy began as workshop craft. The oldest surviving Egyptian documents of the art are the Leiden and Stockholm papyri, of about 250–300 CE. They are recipe books for dyes, imitation gems and imitation gold and silver, with no mysticism in them.

Zosimos of Panopolis, who worked in Egypt around 300 CE, is the earliest alchemical author whose writings survive in quantity. He described furnaces and stills, and framed the work in religious visions; reading them as an inner, psychic transformation is a modern, Jungian interpretation.

English *alchemy* comes through French and Latin from Arabic *al-kimiya*, in which *al-* is the definite article, and the Arabic came from the Greek *khemeia*, a word of unknown origin. One proposal links it to Egypt's name for itself, "the black land".

Alchemists paired the seven metals known since antiquity with the seven classical planets, which included the Sun and the Moon, and gave each metal its planet's sign.

Metal	Planet	Sign
Gold	Sun	☉
Silver	Moon	☽
Quicksilver (mercury)	Mercury	☿
Copper	Venus	♀
Iron	Mars	♂
Tin	Jupiter	♃
Lead	Saturn	♄

Some of the scheme survives in modern words: mercury is the only metal that kept its planet's name, lead poisoning is still called saturnism, and silver nitrate was sold as lunar caustic.

Scholars in the Islamic world gave the art much of its system. Hundreds of treatises survive under the name of Jabir ibn Hayyan, said to have died early in the ninth century, though in 1942–43 Paul Kraus dated them to about 850–950. No source mentions him before about 900, so whether he existed, and when the treatises were written, is still argued over. They hold the oldest known systematic classification of chemical substances.

The Persian physician al-Razi sorted substances into animal, vegetable and mineral, and believed that metals could be transmuted. English still carries the Arabic. *Alembic* comes from *al-inbiq*, from a Greek word for a cup, and *elixir* from *al-iksir*, first a Greek drying powder

for wounds. *Alcohol* comes from *al-kuhl*, the fine stibnite powder used as the eye cosmetic kohl.

Author's note: *If you have ever played a game called Twenty Questions, where one person has to guess a word by asking another person a series of questions, or some variation on that, you have been an alchemist. Your usual first question is something like "Is it alive or not?" — you get a yes or no answer. That next question is really the most interesting one because it captures the essence of this entire book.*

You have so many possible words that could be the answer so you start narrowing it down, or "categorizing things". If alive, you might ask, "Can it move on its own?" If dead, you might ask, "Was it once alive?"

Or your first question might just be "Is it an animal, or plant?" and you have just started playing a game of "make my own alchemy" to try winning the game.

In one version of the game, the answer is first announced as animal, vegetable or mineral, the same three groups al-Razi sorted substances into.

But back to the alchemy. An Arabic book of the late eighth or early ninth century, the *Secret of Creation*, holds the earliest known text of the Emerald Tablet, though its date and origin are disputed. It is also the first known source for the theory that metals grow underground from sulfur and mercury, gold being the best-balanced mixture. That theory made transmutation

sound like adjusting a recipe, and it dominated until the eighteenth century.

In Latin Europe, which read Arabic alchemy in translation from the twelfth century, the most influential text was the *Summa perfectionis*. It was written shortly before 1310 under the borrowed name "Geber"; William Newman suggested in 1985 that its author was a Franciscan, Paul of Taranto. The *Summa* argued from laboratory evidence that metals are made of tiny particles of mercury and sulfur, and the argument reached seventeenth-century chemists, Boyle among them.

The sixteenth-century physician Paracelsus added salt to sulfur and mercury as a third principle, and made remedies from mercury and antimony. "The dose makes the poison", the founding slogan of toxicology, is credited to him. What he wrote, in a defence of 1538, was "Allein die dosis macht das ein ding kein gifft ist": only the dose determines that a thing is not a poison. He was defending his toxic remedies, so the slogan reverses his emphasis.

Robert Boyle's *The Sceptical Chymist* (1661) is often called the book that turned alchemy into chemistry, yet its full title takes aim at "the Generality of Alchymists". Its famous definition of an element, a body "not being made of any other bodies", is offered so that Boyle can doubt that any such body exists. Boyle himself sought the philosophers' stone all his life.

Isaac Newton wrote and copied about a million words on alchemy over some thirty years, and published none of them. They include his English version of the Emerald Tablet, a short, cryptic text credited to Hermes Trismegistus, alchemy's legendary founder. One line reads, in modern spelling:

> That which is below is like that which is above & that which is above is like that which is below to do the miracles of one only thing.

The popular slogan "As above, so below" is not the Tablet's wording. The earliest Arabic says that the upper is from the lower and the lower from the upper. The slogan in its modern form was popularized by, among others, *The Kybalion*, a New Thought tract of 1908. Anyone who has studied the Rider–Waite–Smith deck of tarot cards sees the "as above, so below" more literally drawn out in its Magician, one hand raised towards heaven and the other pointing to the earth.

In 1936 Newton's papers were auctioned, and John Maynard Keynes bought much of the alchemy. In an essay published in 1947, after his death, he wrote:

> Newton was not the first of the age of reason. He was the last of the magicians, the last of the Babylonians and Sumerians, the last great mind which looked out on the visible and intellectual world with

> the same eyes as those who began to build our intellectual inheritance rather less than 10,000 years ago.

It is a fine sentence and a contested one. William Newman's 2018 book on Newton's alchemy treats it as experimental chymistry, continuous with the rest of his science.

Lavoisier completed the turn. His *Traité élémentaire de chimie* (1789) made the conservation of matter chemistry's working principle: "rien ne se crée", nothing is created, by art or by nature. Others had stated the idea before him, but he made it the foundation of quantitative chemistry.

Through all of this, alchemy and chemistry were one subject. Seventeenth-century writers used the two words interchangeably. Restricting *alchemy* to gold-making, as William Newman and Lawrence Principe put it in 1998, "was not widely accepted until the end of the seventeenth century, if even then". Historians now use the old spelling **chymistry** for the single discipline, so as not to read a later split back into it.

Wrong, or right for the wrong reason

Laboratory alchemy worked, some of the time. Its mixing and heating made new substances long before anyone could explain them: mineral acids, distilled alcohol, phosphorus and, by accident, gunpowder and porcelain.

The trouble came with stretching: the same confidence promised gold from base metals and an elixir against death. No chemical reaction can make gold, and no elixir of life has been shown to work.

The most consequential accident began as medicine. Chinese alchemists heated minerals in search of elixirs of long life. The official histories of the Tang dynasty blame elixirs, often made with mercury, lead and arsenic, for the deaths of several ninth-century emperors.

A Daoist text, probably of the mid-ninth century, lists mistakes that elixir-makers should avoid. Some, it warns in Joseph Needham's translation, had heated sulfur, realgar and saltpetre with honey. Smoke and flames followed, "so that their hands and faces have been burnt, and even the whole house (where they were working) burned down".

The honey supplied carbon, so the mixture was a weak gunpowder. Gunpowder incendiaries were probably in use in war by 904, and written formulas appear in a military compendium of 1040–1044. The Chinese word for gunpowder, *huoyao*, still means "fire medicine". A failure in one framework was an invention in another.

In Saxony the search for gold produced something else. Augustus the Strong kept the young alchemist Johann Friedrich Böttger under guard from about 1702 to make him gold. From 1704 Böttger worked under the scientist Ehrenfried Walther von Tschirnhaus.

No gold came. On 15 January 1708 they fired the first European hard-paste porcelain, and the Meissen manufactory was founded on 23 January 1710. Porcelain earned the nickname "white gold". Who deserves the credit for it is still disputed, because Tschirnhaus died in 1708, before Böttger announced the invention.

Böttger survived by delivering a different product. Other gold-makers who took a prince's money and failed were hanged, Georg Honauer at Stuttgart in 1597 and a man known as Cajetani at Küstrin in 1709. They were punished as frauds, not as alchemists, since princes sponsored alchemy widely.

The by-product can be worth more than the goal.

In Hamburg around 1669, Hennig Brand boiled down huge quantities of urine in search of the philosophers' stone, then heated the residue strongly. He got a waxy substance that glowed in the dark and burst into flame: phosphorus, the first element with a known discoverer.

Brand kept his method secret and sold it for 200 thalers. Johann Kunckel worked it out again by 1678, and in 1680 Robert Boyle did so too and published a method. A wrong theory, followed carefully, had found something real, and secrecy slowed everyone else down.

The central goal of alchemy turned out to be possible after all. In 1901 and 1902, at McGill University, Ernest Rutherford and Frederick Soddy showed that thorium's atoms change into atoms of other elements.

Soddy later recalled telling Rutherford that this was transmutation, and Rutherford replying: "For Mike's sake, Soddy, don't call it transmutation. They'll have our heads off as alchemists." Their 1902 papers said "transformation". Thirty-five years later, Rutherford called a book *The Newer Alchemy* (1937).

Enrico Fermi went further. Bombarding uranium with neutrons, he suggested in 1934 that he might have made a new element, number 93 or higher; by 1938 he claimed two, elements 93 and 94, named ausenium and hesperium.

These "new elements" were part of the work cited for his Nobel Prize in Physics in 1938. Within weeks of the ceremony, the discovery of fission showed that they were fragments of split uranium, and Fermi added a footnote to his published Nobel lecture. He had transmuted uranium, though not into what he thought.

Gold itself followed. In 1980 Glenn Seaborg's team at Berkeley fired atomic nuclei at bismuth-209 in a particle accelerator and made a tiny amount of gold. Seaborg put the cost at more than a quadrillion dollars an ounce.

In 2025 CERN reported that near-miss collisions of lead nuclei in the Large Hadron Collider had made about 86 billion gold nuclei between 2015 and 2018, 29 picograms in all. Each one broke apart against the machine a fraction of a second after it formed.

The goal was physically possible. It needed nuclear physics rather than chemistry, and it was worthless

once achieved. One more case, Carnot's, belongs with the story of heat that follows.

Thermodynamics for a ten-year-old

The word **thermodynamics**, the science of heat and work, arrived late, after most of what it names had been found. Told one picture at a time, as for a ten-year-old, its story begins with an alchemist.

Block the end of a bicycle pump and push the handle. The trapped air pushes back, and squeezed into half the space it pushes twice as hard. Robert Boyle published that rule in 1662, as tables of measurements rather than an equation, and the same Boyle hoped all his life to make gold.

Jacques Charles noticed around 1787 that different gases swell by equal amounts when heated, and John Dalton and Joseph Louis Gay-Lussac both published measurements of the rule in 1802. A hot-air balloon ride shows this in action; a video works for anyone who would rather keep their feet on the ground.

Émile Clapeyron combined these rules into one equation in 1834, and in the 1870s chemists made it universal, the same for every gas. Written today as $PV = nRT$, it says that a gas's pressure times its volume depends only on how much gas there is and how hot it is.

A separate thread began with an engine built without any theory of heat. In 1816 Robert Stirling, a

Church of Scotland minister ordained eight days earlier, applied for a patent for “Improvements for diminishing the consumption of fuel”.

His engine shuttled the same air between a hot end and a cold end through a mass of metal that he called the economiser. It worked like a heat sponge, soaking up heat on the way to the cold side and giving it back on the return. A safety advantage over exploding steam boilers was claimed for the engine by the 1840s, but the 1816 patent itself was about saving fuel.

Sadi Carnot asked how much work heat can give. His book of 1824 pictured heat as caloric, an invisible fluid that falls from a hot body to a cold one as water falls onto a mill wheel, without being used up.

Here the accepted idea of moving water did real work, but the picture was wrong: part of the heat leaving a furnace becomes the work itself. Carnot’s conclusion survived anyway. The best any engine can do depends only on the temperatures of its hot and cold sides, not on whether it works with steam or air. A bigger drop from hot to cold gives more work, as a higher waterfall does.

Carnot’s private notes, printed in 1878, show that he had given up caloric by 1832. In 1850 Rudolf Clausius kept Carnot’s result while replacing caloric with the conservation of energy. In the ideal case, an engine like Stirling’s with a perfect heat sponge reaches exactly Carnot’s limit.

James Joule supplied the right reason in the 1840s: heat and work turn into each other at a fixed rate. A falling weight that churns paddles in water warms the water, and rubbing hands together warms them for the same reason.

Only then did the subject get its name. William Thomson wrote of a "thermo-dynamic engine" in 1849 and of "thermo-dynamics" in 1854, and William Rankine's manual of 1859 used the closed form, "thermodynamics". Clausius coined "entropy" in 1865. The name came last, after the engines and the measurements, and once it existed the ideas could travel.

Alchemy turned inward

Alchemy always spoke a religious language, but the reading in which it was never about metals, only about the soul, is mostly modern. Mary Anne Atwood argued it in 1850 and Ethan Allen Hitchcock in 1857; Herbert Silberer gave alchemy a psychoanalytic reading in 1914, before Carl Jung made the inward reading famous. Lawrence Principe and William Newman, whose view prevails among historians of science, argued in 2001 that a purely spiritual alchemy is a nineteenth-century construction. Historians of esotericism such as George-Florin Călian answered in 2010 that some older texts really are spiritual. Daoist inner alchemy, *neidan*, is an older exception, in a tradition of its own.

The language had already drifted towards allegory in 1614–16, when three anonymous texts announced a secret Rosicrucian brotherhood, for which there is no evidence. The third, *The Chymical Wedding of Christian Rosenkreutz*, is an allegorical romance about a royal "chymical", or alchemical, wedding. Its author, the Lutheran theologian Johann Valentin Andreae, later called it a *ludibrium*, a jest.

The modern revival came through ritual magic. The Hermetic Order of the Golden Dawn, founded in London in 1888, admitted women on equal terms and taught alchemy in its inner order alongside ritual magic. In 1898 Aleister Crowley joined, introduced by a chemist with whom he had been discussing alchemy, and within two years the order split.

In 1904 Crowley recast the demon-summoning of the Goetia as psychology: "The spirits of the Goetia are portions of the human brain." His tarot, painted by Frieda Harris between 1938 and 1943, turned the Temperance card into "Art", a figure mixing fire and water beneath a Latin motto whose initials spell VITRIOL. Beyond the Golden Dawn, that card and that essay, his ties to alchemy are thin.

Jung came to alchemy through a Daoist inner-alchemy text, *The Secret of the Golden Flower*, which Richard Wilhelm translated in 1929 with Jung's commentary. Wilhelm thought it carried Tang-dynasty

teaching, but later scholars traced it to spirit-writing groups of 1688 and 1692.

In *Psychologie und Alchemie* (1944) Jung argued that alchemists, not knowing what matter was, projected their own unconscious onto it. On this reading, the stages of the work picture individuation, a person's growth towards wholeness. Historians of alchemy fault him for lifting it out of its history, but his influence on popular psychology and the New Age movement was immense.

The historian Wouter Hanegraaff, writing in 1996, called the wider trend the "psychologizing of religion and sacralization of psychology". One strand reached Alcoholics Anonymous, of which Jung is often called a godparent.

In 1961 Jung wrote to Bill Wilson, a co-founder, that the Latin for alcohol is *spiritus*, and closed with the formula *spiritus contra spiritum*: spirit against spirit. The pun has an alchemical root. The "spirits" of Arabic alchemy were the volatile substances that fly off in the fire, and distilled drink, by the usual dictionary account, inherited the word.

The best-known modern use of alchemy as self-transformation is a novel. Paulo Coelho's *O Alquimista* appeared in Brazil in 1988, in a first edition that sold only about 900 copies. Alan R. Clarke's English translation, *The Alchemist*, followed in 1993.

An Andalusian shepherd boy dreams of treasure at the Egyptian pyramids and sets out to find it. He meets an Englishman who studies books in search of an alchemist, and then the alchemist himself, who teaches by experience. At the pyramids a mocking thief lets slip that the treasure lies at home in Spain, the twist of an old folktale, told by Rumi and in the *Thousand and One Nights*.

In Clarke's translation, the book's central maxim reads:

> When you want something, all the universe conspires in helping you to achieve it.

The novel has sold tens of millions of copies. Critics read it as a spiritual fable: *Publishers Weekly* called it "a simple fable" in 2003, and Gregory Cowles of *The New York Times* called it "more self-help than literature" in 2009.

The biologist Richard Dawkins coined the word **meme** in 1976, in *The Selfish Gene*, for a unit of culture, such as a tune, a catchphrase or a fashion, that spreads by being copied from one mind to another as genes are copied from parent to offspring.

A comment from the author, not a finding: alchemy's own words have travelled like memes. "Spirits" passed from the furnace to the bottle, and "alchemy" now names almost any mysterious change, from a chef's to a novelist's. Each copy changed the word a little, which is how

memes travel, and why the idea works better as shorthand than as a mechanism.

What follows is the author's reading of what the inward alchemy was for.

I read much of this later alchemy as a way of building a person. It names the parts of a character, balances them, and works towards an alignment, much as the laboratory texts balanced sulfur against mercury. Often it is self-help in strange words.

It was productive in ways it did not always recognize. It gave people stages to move through, a vocabulary for change, and permission to treat the self as something that can be worked on. Coelho's shepherd is that reading turned into a story.

That is the purpose I think esotericism can serve. Its value is psychological, and it is no evidence about matter. Whether it reaches anything beyond the psychological is a question this primer leaves open.

A shelf of alchemical books

There was never an official alchemical canon. These ten are among the works most printed, copied and cited, oldest first.

- The Emerald Tablet, in Arabic by the early ninth century and in Latin from the twelfth: a short, cryptic text credited to the legendary Hermes Trismegistus.

- The *Summa perfectionis magisterii*, written in Latin shortly before 1310 under the borrowed name "Geber": the clearest medieval laboratory manual.
- *The Compound of Alchymy*, English verse dated 1471 by its colophon, in which George Ripley walks through twelve "gates", from calcination to projection.
- *Splendor Solis*, whose oldest copy is dated 1531/32: one of the most famous illuminated alchemical manuscripts, by an unknown author.
- The *Rosarium philosophorum*, printed at Frankfurt in 1550: a "rose garden" of sayings with twenty woodcuts, a series of which Jung later read as pictures of the psyche.
- *The Twelve Keys of Basil Valentine*, 1599: allegories that decode as metallurgy, under the name of a monk invented after the texts appeared.
- Michael Maier's *Atalanta fugiens*, 1617: fifty emblems, each with an engraving, verses, a short essay and a three-voice fugue.
- The *Mutus liber*, La Rochelle, 1677: fifteen engraved plates and no explanatory text, secrecy taken to its limit.
- Isaac Newton's alchemical manuscripts: about a million words, which he never published, dispersed at auction in 1936.
- Carl Jung's *Psychologie und Alchemie*, 1944: the modern inward reading, which most historians of

science reject as a history of alchemy and historians of esotericism defend.

Two famous modern books borrow alchemy's language without belonging on that shelf. Aleister Crowley's *777* (1909) is a reference book of Qabalistic correspondence tables for ceremonial magic, much of it reworked from the Golden Dawn's; of its 183 columns, two are alchemical. Coelho's *The Alchemist* (1988) is a novel that uses alchemy as a metaphor.

Handles and gates

A name gives the mind something to hold. Once things had names, chemists could carry ideas from one problem to another and combine them into new ones.

Lavoisier saw this. In 1787 he and three colleagues replaced descriptive names such as "flowers of zinc" and "oil of vitriol" with systematic ones such as "copper sulfate". His preface to the *Traité*, in Robert Kerr's translation of 1790, gives the reason:

> We cannot improve the language of any science without at the same time improving the science itself; neither can we, on the other hand, improve a science, without improving the language or nomenclature which belongs to it.

The same preface quotes the philosopher Condillac: "We think only through the medium of words."

Alchemists had often done the opposite, on purpose. Historians call their code words **Decknamen**, German for "cover names". In the first of Basil Valentine's *Twelve Keys*, a fierce grey wolf devours the body of the King and is burnt to ashes, three times over. Decoded, the wolf is stibnite, an antimony ore, the King is gold, and the passage is a way of refining gold.

Boyle complained in 1661 that chymists wrote in an "obscure, Ambiguous, and almost AEnigmatical Way", to be understood only "by the Sons of Art". He suspected that some wrote confusedly because they understood confusedly. William Newman argued in 1996 that an obscure style also showed that a writer belonged to the fraternity of adepts; otherwise "he would not be taken seriously".

Not all of it was gatekeeping. Many cover names were ordinary technical shorthand. Some secrecy was caution: in 1676 Newton urged that Boyle keep "high silence" about a mercury that grew hot with gold, in case making it known did "immense dammage to the world". Some was legal, since an English statute of 1404 made multiplying gold or silver a felony.

The codes could also be broken. By the mid-1670s a Leiden reader, Carolus de Maets, had worked out the process hidden in the allegories of George Starkey, who wrote as the adept "Philalethes".

A comment from the author, not a finding: modern fields do something similar, mostly without meaning to. A

phrase that makes sense inside a field, or inside any in-group, shuts out everyone who has not first learnt the whole field. Academic gatekeeping often follows, from the nature of the language rather than from anyone's intent. Some of that is necessary shorthand, as many cover names were. The rest works as a gate, and Lavoisier's remedy still seems the right one to me.

What this book is

What follows is the author's reading of what this book is for.

So that is what this book is: a way to think about fields that have already been studied, and a set of analogies for understanding how and why the world works the way it does. The alchemists borrowed the Sun, the Moon and moving water because those were the ideas they had. The glossary that follows offers newer ones, each with a symbol, and a notation for combining them.

My simplifications are where the checking starts, and the corrections they draw out usually make the story more interesting. My first notes for this primer said alchemy promised what physics forbids. Physics allows it: Seaborg's team made gold in 1980, and in 2025 CERN reported gold made in its collider. The same notes said most plants cannot grow in the dark; they can, tall and pale, and then they starve. Both corrections made better stories.

Anyone who wants to can use what follows to build new theories, or to start a new form of alchemy. It begins with what its own simplifications cost. I'm not greedy, just curious and wanting to share my findings.

Sources for the primer

These are the works the primer relies on, in the author–date form used in *Natural Knowledge* and in alphabetical order. An entry marked † has confirmed details but was not checked against the work itself, because the supporting passage was found in a secondary account or a catalogue record; treat the † as a pointer to where to look. Some settled points, such as the origins of words, rest on encyclopedia summaries and have no entry here.

Reading further

Lawrence Principe's *The Secrets of Alchemy* (2013) is a readable history of the whole art, by a historian who has reconstructed alchemical recipes in the laboratory.

Robert Boyle's *The Sceptical Chymist* (1661) and **Antoine Lavoisier**'s *Elements of Chemistry*, in Robert Kerr's translation of 1790, are both free on Project Gutenberg.

Sadi Carnot's *Reflections on the Motive Power of Heat*, in the 1897 edition of R. H. Thurston's translation (first published 1890), is short and free online.

The Chymistry of Isaac Newton, from Indiana University, puts Newton's alchemical manuscripts online, with a glossary of the alchemists' cover names.

Furnace and Fugue (2020) is a digital edition of Michael Maier's *Atalanta fugiens*, with scholarly commentary.

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This is oversimplified. On purpose.

Every entry here compresses a field into one line. "Autocatalysis — a reaction product catalyses the production of more of itself" throws away the kinetics, the thermodynamic constraints, the decades of argument about whether autocatalytic sets can actually bootstrap under plausible prebiotic conditions, and most of what a chemist means by the word.

That loss is the entire point, and it is also the entire danger.

The point: you cannot hold a field in your head, but you can hold a handle. Handles let you ask "what happens if I put *this* next to *that*" without first earning a doctorate in both. Most useful cross-domain questions start as exactly that kind of naive juxtaposition.

The danger: a handle feels like understanding. Writing `Au + C → protocell stability` produces a sensation of insight that is not insight. The symbols combine cleanly because symbols always do. The chemistry does not care.

So: this is a starting point, not a result. If a combination here suggests something interesting, that is the

beginning of reading the actual literature, not a substitute for it. The notation cannot be wrong, which means it cannot be right either — only the claims you make with it can be.

This is also why the name is a joke with a point in it. Alchemy had a real notation, real apparatus, and a genuine conviction that the right combinations would reveal how matter worked. It was wrong about nearly everything, and it was still the thing chemistry grew out of. Being a useful wrong starting point is not nothing.

The ID scheme

Every concept has a permanent ID: a two-letter tier prefix and a number.

`CO-01 Variation` — tier CO = core process vocabulary

`PH-04 Brownian motion` — tier PH = physical foundations

`PL-07 Ribozymes` — tier PL = proto-life

Tiers:

Prefix	Tier
`CO`	Core process vocabulary — the engine the rest runs on
`PH`	Physical foundations
`CH`	Chemical emergence
`PL`	Proto-life
`BI`	Biological evolution
`CX`	Complex systems and information
`MC`	Mind, language, and culture
`MF`	Meta-frame — cross-domain framings, the most speculative tier

Rules, so references don't rot:

1. A new term takes the next free number in its tier. Numbers are never reused and never reassigned.
2. If a term is withdrawn, its ID is retired, not recycled. The entry stays with a note saying what replaced it.
3. If a term moves tiers, it keeps its original ID and gains a cross-reference. Renumbering breaks every citation anyone has made.
4. One symbol per concept, repo-wide. Collisions are bugs.

Why this replaces the original riff

The original riff, an early unpublished draft of this vocabulary, is the seed for all of this, and its stated purpose was to make concepts referenceable. Two defects prevented that:

- **It had two numbered lists.** A Master Glossary numbered 1–56 and an Ontology Tree numbered 1–55, where the same number meant different things. `#10` was *Autocatalysis* in one and *Catalysis* in the other. A reference to "#10" was ambiguous.
- `G` **was assigned twice** in the Symbolic Glossary — "concentration gradients" and "gene-centered evolution."

The second is fixed by assignment: G is concentration gradients, as the original equations used it; gene-centred evolution becomes Ge.

The first is fixed structurally. Having one list and one hierarchy was the bug — the tiers in the glossary *are* the ontology, so there is no second list to fall out of sync. The riff is kept as history; use the glossary for anything live.

Glossary

The canonical concept list. Every term has a permanent ID, a unique symbol, and a one-line gloss.

The tier groupings below *are* the ontology. There is no second hierarchy to fall out of sync with this one — that was the defect in the original riff, and it is fixed structurally rather than by maintenance.

Read "This is oversimplified" and "The ID scheme" first. Every gloss here is a drastic simplification, and the ID rules matter if you intend to cite anything.

Roughly ordered from "things physics does" to "things people do." That ordering is a convenience for finding terms, not a claim that the later tiers reduce to the earlier ones.

CO — Core process vocabulary

The engine. These recur in every tier below, which is the whole observation the book is built on. Also the tier where the book is most insistent that a shared word is not a shared mechanism.

ID	Term	Sym	Gloss
CO-01	Variation	`Va`	The production of differences within a system.
CO-02	Search	`Sr`	Exploration of possible states. Needs no intent, intelligence, or goal.
CO-03	Selection	`Sl`	Differential persistence or success of some configurations over others.
CO-04	Retention	`Rt`	Persistence of a configuration over time.
CO-05	Amplification	`Am`	A pattern becoming more abundant or more influential.
CO-06	Specialization	`Sp`	Components developing distinct functions. Buys efficiency, costs independence.
CO-07	Compression	`Cm`	Reduction of complexity into a more manageable form. Always lossy.
CO-08	Transmission	`Tx`	Movement of a pattern between components, organisms, or generations.
CO-09	Emergence	`Em`	System-level properties arising from interaction among components.
CO-10	Feedback	`Fb`	Output influencing subsequent behaviour. Stabilising or destabilising.
CO-11	Coupling	`Cu`	Linking of processes so one supports, regulates, or enables another.

ID	Term	Sym	Gloss
CO-12	Accumulation	Ac	Changes retained and layered over time.
CO-13	Selection bias	Sb	Observing what persisted rather than everything that failed.
CO-14	Concentration	Cn	Local accumulation of matter, energy, or information.
CO-15	Compartmentalization	C	A boundary that localizes processes and permits distinct internal conditions.
CO-16	Metacompartmentalization	MCp	Compartments nested within, or integrated into, larger compartments.

PH — Physical foundations

ID	Term	Sym	Gloss
PH-01	Thermodynamics	T	Constraints on energy, work, and order.
PH-02	Entropy	S	Measure of dispersal; the direction isolated systems drift.
PH-03	Energy gradient	Eg	A difference in energy that can be exploited to do work.
PH-04	Concentration gradient	G	A difference in concentration across a distance or boundary.
PH-05	Electrochemical gradient	Eq	Combined concentration and charge difference across a membrane.
PH-06	Non-linear dynamics	N	Small causes producing disproportionate effects.

ID	Term	Sym	Gloss
PH-07	Chaos	Ch	Sensitive dependence on initial conditions. Deterministic, not random.
PH-08	Brownian motion	B	Random thermal motion of particles. The stirring that enables encounters.
PH-09	Diffusion	Df	Net movement down a concentration gradient.
PH-10	Cyclic forcing	Cy	Repeating environmental rhythms — tides, day/night, wet/dry, seasons.
PH-11	Percolation threshold	P	The density at which connectivity appears suddenly across a network.
PH-12	Fractals / self-similarity	F	Structure repeating across scales.
PH-13	Scaling	Scl	How properties change with size.
PH-14	Modularity	Mo	Composition from semi-independent, recombinable units.
PH-15	Dissipative structures	DS	Order sustained by continuous energy throughput. Stop the flow, lose the form.
PH-16	Electromagnetic radiation	EMr	Self-propagating oscillation of electric and magnetic fields.

CH — Chemical emergence

ID	Term	Sym	Gloss
CH-01	Catalysis	Ca	A substance accelerating a reaction without net consumption.

ID	Term	Sym	Gloss
CH-02	Autocatalysis	`Au`	A product catalysing production of more of itself.
CH-03	Autocatalytic set	`AS`	A reaction network whose members collectively sustain the network.
CH-04	Reaction network	`RN`	Connected reactions sharing products and reactants.
CH-05	Reaction–diffusion	`RD`	Pattern formation from reaction coupled to diffusion.
CH-06	Belousov–Zhabotinsky reaction	`BZ`	A chemical oscillator. Demonstrates order without life.
CH-07	Prebiotic chemistry	`Pre`	Chemistry of the early Earth, before biology.
CH-08	Miller–Urey synthesis	`MU`	Amino acids from simple gases plus an energy source.
CH-09	Spark-discharge chemistry	`SD`	The wider class of energy-driven prebiotic synthesis.
CH-10	Hydrothermal vent gradients	`HV`	Natural thermal, pH, and redox gradients at ocean-floor vents.
CH-11	Iron–sulfur world	`IS`	Metabolism-first hypothesis using mineral-surface catalysis.
CH-12	Amphiphile	`Ap`	A molecule with water-attracting and water-repelling ends.
CH-13	Chemical oscillation	`Os`	A reaction system cycling rather than settling to equilibrium.

PL — Proto-life

ID	Term	Sym	Gloss
PL-01	Lipid vesicle	V	A spontaneously forming bilayer sac enclosing a volume.
PL-02	Micelle	Mi	An amphiphile cluster forming a boundary. Forms unaided.
PL-03	Membrane	Mb	A boundary that separates and selectively permits exchange.
PL-04	Protocell	Pc	A compartment containing self-maintaining chemistry. Not yet alive.
PL-05	RNA world	R	Hypothesis that RNA once carried both information and catalysis.
PL-06	Ribozyme	Rb	Catalytic RNA. Information and function in one molecule.
PL-07	Template replication	TR	Copying by structural matching.
PL-08	Hypercycle	Hc	Eigen's cooperative network of mutually supporting replicators.
PL-09	Metabolism	M	Chemistry that maintains the system running it.
PL-10	Heredity	H	Retention and copying of structure across generations.
PL-11	Chemiosmosis	Pg	Using a proton gradient across a membrane to drive work.
PL-12	ATP	ATP	The molecule cells use to move usable energy around.

BI — Biological evolution

ID	Term	Sym	Gloss
BI-01	Natural selection	NS	Differential reproduction of heritable variants.
BI-02	Mutation	Mt	Heritable change in genetic sequence.
BI-03	Genetic drift	Dr	Allele frequency change by sampling chance, not fitness.
BI-04	Gene flow	Gf	Movement of genetic material between populations.
BI-05	Adaptation	Ad	A trait improving fit under particular conditions. Not under all.
BI-06	Gene-centred evolution	Ge	Treating the gene as the unit selection acts on.
BI-07	Selfish gene	SG	Dawkins' framing: genes as the persisting replicators.
BI-08	Endosymbiosis	En	One organism becoming a permanent component of another.
BI-09	Multicellularity	Mx	Cells specializing and persisting as a coordinated organism.
BI-10	Symbiosis	Sy	Persistent close association between different organisms.
BI-11	Microbiome	Mm	The microbial community associated with a host or environment.
BI-12	Holobiont	Hb	Host plus associated microbes, treated as one system. Use with care.
BI-13	Distributed labour	DL	Function spread across components rather than held by one.

ID	Term	Sym	Gloss
BI-14	Convergent evolution	Cnv	Unrelated lineages arriving at similar solutions under similar constraints.
BI-15	Vestigial structure	Vs	A structure retained past its function. Evidence of path dependence.
BI-16	Horizontal gene transfer	HGT	Genetic material moving between organisms outside reproduction.
BI-17	Homeostasis	Ho	Active maintenance of internal conditions against external change.
BI-18	Ecological network	Ec	The web of interactions among organisms sharing an environment.

CX — Complex systems and information

ID	Term	Sym	Gloss
CX-01	Information theory	I	Formal treatment of storage, transmission, noise, and compression.
CX-02	Network theory	Ne	How connectivity produces robustness or fragility.
CX-03	Robustness	Rs	Continued function despite perturbation or component loss.
CX-04	Self-organization	SO	Global order arising from local rules, with no controller.
CX-05	Interdependence	Id	Components relying on one another. Specialization's standing cost.
CX-06	Cellular automata	Aut	Grids of cells following local rules. Toy models of emergence.

ID	Term	Sym	Gloss
CX-07	Conway's Game of Life	GL	Four rules producing gliders, oscillators, and replicators.
CX-08	Agent-based model	ABM	Many simple units simulated to observe collective behaviour.
CX-09	Error correction	Er	Using redundancy to survive corruption.
CX-10	Search space	SS	The set of configurations a system could occupy.
CX-11	Combinatorial explosion	Cbx	Possibility counts growing faster than any search can cover.

MC — Mind, language, and culture

ID	Term	Sym	Gloss
MC-01	Pattern recognition	PR	Detecting regularity in input. Powerful and unreliable in the same breath.
MC-02	Memory	Mr	Retention of information for later use.
MC-03	Prediction	Pd	Using a model to anticipate what comes next.
MC-04	Categorization	Cg	Grouping distinct inputs as equivalent for some purpose.
MC-05	Abstraction	Ab	Discarding detail to retain what generalizes.
MC-06	Representation	Rp	An internal stand-in for something external.
MC-07	Strange loop	Stl	A system complex enough to model its own modelling. Hofstadter.

ID	Term	Sym	Gloss
MC-08	Cognitive constraint	CC	Limits of attention, memory, and vocabulary shaping what can be thought.
MC-09	Language evolution	L	Change in a language through use, variation, selection, and transmission.
MC-10	Phoneme	Ph	A sound unit a language treats as distinct.
MC-11	Articulatory constraint	Art	Limits of the vocal tract shaping which sounds recur across languages.
MC-12	Semiotics	Sm	The study of signs and meaning.
MC-13	Symbolic compression	SC	A symbol standing for an object, event, or relation.
MC-14	Grammar	Gr	Constraints on how units may combine.
MC-15	Cultural transmission	CT	Movement of practices and ideas between people and generations.
MC-16	Meme selection	Me	Differential persistence of cultural units. Fidelity is far lower than genetic.
MC-17	Cultural bottleneck	CB	Isolation or catastrophe narrowing what survives.
MC-18	Founder effect	FE	A small starting group disproportionately shaping what follows.
MC-19	In-group / outgroup dynamics	UT	Group boundaries driving divergence and cohesion together.

ID	Term	Sym	Gloss
MC-20	Economic memetics	Ek	Economic practices persisting because they work under local conditions.
MC-21	Land-use evolution	LU	Change in land practice under ecological and social selection.
MC-22	Myth as explanation	My	Narrative filling a causal gap when mechanism is unavailable.
MC-23	Religious evolution	Re	Change in belief systems under social and cognitive selection.
MC-24	Explanatory uncertainty	EU	The condition under which explanatory systems are most readily adopted.

MF — Meta-frame

The most speculative tier. Everything here is a framing the author finds useful, not an established result. Nothing in this tier should be used as a premise without saying out loud that it is one.

ID	Term	Sym	Gloss
MF-01	Life as search-space explosion	LS	Early Earth as a massively parallel chemical trial, read through survivor bias.
MF-02	Language as protochemistry	LaC	Sound units combining like reactive units. **Analogy, not mechanism.**
MF-03	Religion as cultural selection	RC	Belief systems persisting by fit to minds and societies. Says nothing about truth.

ID	Term	Sym	Gloss
MF-04	Life as a cluster of properties	LC	"Alive" as a family resemblance, not a sharp boundary.
MF-05	Culture as cumulative information	CI	Culture as information outliving the individuals carrying it.
MF-06	Cross-domain continuity	XD	The claim that one vocabulary spans these domains. The book's actual thesis.
MF-07	Spiral as the geometry of becoming	GB	Spirals recurring where systems expand while retaining continuity. Contested.
MF-08	Time as repetition plus accumulation	TA	Cycles that never close, because each turn happens in a changed system.
MF-09	Why-are-we-here inquiry	WH	Self-aware systems asking after their own origin.

Retired IDs

None yet. When a term is withdrawn, its row stays here with a note on what replaced it. IDs are never recycled.

Changes from the original riff

The riff is an early unpublished draft of this vocabulary. For anyone reconciling against it:

- **Two numbered lists merged into one.** The Master Glossary (1–56) and Ontology Tree (1–55) disagreed about what each number meant.

- `G` **collision resolved.** `G` is PH-04 concentration gradient, matching the original equations. Gene-centred evolution becomes `Ge` (BI-06).
- **Symbols made unique case-insensitively.** `Sl`/`SL`, `Ac`/`AC`, `Sc`/`SC`, `Cb`/`CB`, `Ca`/`CA`, `Mu`/`MU`, `Em`/`EM`, `Pc`/`PC` would all have collided in a case-insensitive lookup. Renamed: strange loop `Stl`, articulatory constraint `Art`, scaling `Scl`, combinatorial explosion `Cbx`, cellular automata `Aut`, mutation `Mt`, electromagnetic radiation `EMr`, prebiotic chemistry `Pre`.
- **Dropped:** "dad/mom sound recurrence" (an example, not a concept — it belongs in the book's chapter 14), and "unknown-force hypothesis" (a placeholder for absent knowledge, which is not a concept you can compose with).
- **Added** from the book's own vocabulary: the whole CO tier as a named group, plus entropy, diffusion, amphiphile, protocell, chemiosmosis, ATP, mutation, drift, gene flow, endosymbiosis, microbiome, holobiont, convergent evolution, vestigial structure, horizontal gene transfer, homeostasis, robustness, error correction, and the MC cognition terms.

Notation

A shorthand for combining the concepts in the glossary.

Read the last two sections before you use any of this seriously. The notation's limits are more interesting than its operators, and they are where the honest work is.

Operators

Op	Reads as	Means
+	"and"	These act together. Order carries no meaning.
→	"can produce"	The left side can give rise to the right under suitable conditions. **Not** "always does."
↺	"cycles with"	Mutual reinforcement; output feeds back to input.
⊕	"interacts with"	They affect each other, with no claim about direction.
↑	"amplifies"	The left increases the magnitude or reach of the right.
↓	"constrains"	The left limits what the right can do.
≈	"is analogous to"	Structural resemblance. **Explicitly not a mechanism claim.**
⊂	"is a kind of"	The left is a special case of the right.

Op	Reads as	Means
‖	"runs parallel to"	Both proceed at once, independently.

Two of these carry nearly all the risk. → invites reading as "causes," and ≈ invites reading as "is the same as." Neither is meant that way. If a sentence you write from a formula would be false with "can sometimes produce, under conditions not specified here" substituted for →, the formula is overstating.

Templates

x + y → z — two things together can produce a third
x ↺ y — mutual reinforcement
x ≈ y — structural analogy, no shared mechanism claimed
x ⊂ y — x is a special case of y
x ↓ y — x limits y

Worked chains

Chemistry gets started

T + Cy + Ca → Sr — energy, rhythm, and catalysis make a search
Pre + G + Cy + Ca → RN — that search produces reaction networks
Au + RN → AS — some networks start sustaining themselves

`MU + SD → Pre` — and the feedstock has a demonstrated route

`T + Eg + Cy → Pre` — energy flowing through a cycling world drives early synthesis

Chemistry gets a boundary

`Pre + RN → Ap` — the chemistry that builds networks also yields amphiphiles

`Ap + Df → Mi` — amphiphiles in water cluster unaided

`Ap + Df → V` — and, at the right concentration and acidity, close into a sac

`V + Cn → C` — an enclosed volume makes concentration local

`AS + C → Pc` — self-sustaining chemistry, now enclosed

`Mb + Eq → Pg` — a maintained gradient becomes usable energy

`HV + IS + G → Pg` — one plausible natural route to the same thing

Chemistry gets a memory

`R + TR + Va → H` — copying with error is heredity

`Pc + H → NS` — a protocell that takes a copier in becomes what selection acts on

`H + Va + Sl → NS` — heredity under selection is evolution

`NS + Ac → Ad` — selection plus time is adaptation

Rb ⊂ R — ribozymes are the RNA world's working case

Organization stacks

C + Sp → DL — boundaries plus specialization distribute labour

DL + Cu → MCp — coordinated specialists are a larger compartment

NS + Sp + Cu → Mx — selection on cooperating, specialised cells makes a body

Mx ⊂ MCp — a multicellular body is nesting's most familiar case

En ⊂ MCp — endosymbiosis is metacompartmentalization's sharpest case

Mm + Ec + Fb → Hb — host and microbes as one interacting system

Sp ↑ Id — every gain in specialization costs independence

Minds and culture

Mx + Sp + NS → PR — selection on specialised cells in a body makes pattern recognition

PR + Mr + Pd → Rp — pattern, memory, and anticipation make a model

Rp + Cm → Ab — compressing the model is abstraction

Rp ↺ Rp → Stl — a model that models its own modelling

Stl + SC → L — externalising the loop needs symbols

L + Gr + CT → CI — language plus transmission accumulates culture

CC + EU + My → Re — constraint, uncertainty, and narrative

CB + FE + UT ↑ Va — isolation increases cultural divergence

Art ↓ Ph — anatomy limits which sounds recur

The whole arc

T → Pre → RN → C → Pc → NS → Mx → Stl → L → CT → Re

RN → H → NS

The arc has two strands, and they meet at NS. Along the first, chemistry closes into compartments. Along the second comes copying, which joins when a protocell takes a copier in. That join is the author's reading in chapter 20, of the first cell as a combination rather than a transformation, and Pc + H → NS above is the step. Each arrow here compresses one or more chains above, and the fact checker in *Prolog Logicbase*, book 4 of this collection, tests that every one does. Two kinds of step have no chain, and the book says so itself. RN → H is the gap chapter 8 cannot close: how chemistry first copied itself reliably is not known. Giving copying its own strand changes where that gap sits, not whether it is open. And L → CT → Re sits above language, where

chapter 16 calls the continuity shared vocabulary rather than shared mechanism.

Compressed to one line: LS ≈ T + B + Cy + Ca + G + Sb. Early Earth as a parallel chemical trial, read through survivor bias.

Analogies, flagged as analogies

LaC : L ≈ Pre — language assembles units the way chemistry does

RC : Re ≈ NS — beliefs persist by fit, as organisms do

Me ≈ Ge — cultural and genetic replicators — far lower fidelity

These are ≈, never →. They are the book's most-criticized move, and writing them with the right operator is the cheapest honesty available.

What this notation cannot say

The important section. Every line below is something a formula silently omits, which means every formula is more confident than the thing it describes.

Rates and timescales. Pre → RN holds whether it took an hour or a hundred million years. Most of the real difficulty in origin-of-life research is timescales, and none of it survives the arrow.

Probability. → does not distinguish "reliably" from "once, somewhere, in a planet's worth of trials." For abiogenesis that distinction is the entire question.

Necessity versus sufficiency. `x + y → z` does not say whether both are required, whether either alone would do, or whether both together are merely sometimes enough. Three very different claims, one notation.

Conditions. Everything here is implicitly "under suitable conditions," and suitable conditions are where the science lives.

Scale. `C` covers a micelle and a multicellular organism. Useful for seeing the pattern; useless for saying anything true about either.

Reversibility. Nothing marks which arrows run backwards. Specialization reverses. Compartments dissolve. Languages lose distinctions.

Failure. Only what persisted gets written down — which is `Sb`, selection bias, appearing inside the notation's own design. Worth sitting with.

Substrate. The sharpest limit. `S1` appears in chemical, biological, and cultural chains as if it were one thing. It is not. Chemical persistence, differential reproduction, and an idea being repeated are three unrelated mechanisms wearing one word. The notation makes them look identical because notation cannot see mechanism. Whenever a chain crosses tiers, the real work starts exactly where the formula stops.

Using this without fooling yourself

1. **A formula is a question, not a finding.** `x + y → z` means "is this worth checking?"
2. **Crossing a tier boundary demands prose.** A chain from `CH` to `MC` has smuggled in an analogy. Name it.
3. **If it composes too easily, suspect it.** Symbols always combine. That is a property of symbols.
4. **Write the sentence.** Any formula you cannot state as a plain English claim, with its conditions attached, is not saying anything yet.
5. **Cite the glossary ID, not the symbol,** in anything durable. `BI-06` is unambiguous forever; `Ge` is unambiguous only while this book is open.

The original riff ended by offering to turn all of this into "a compact set of equations" expressing the whole theory. That offer is worth declining. A theory you can write as equations with no empirical content is not compact — it is empty. What this notation is good for is finding the places where two domains seem to rhyme, so you can go and find out whether they actually do.

About the author

Freddi Van Gogh grew up moving between England, Germany, and the United States, often left to themselves, out in nature or in an empty Oxbridge library, with a great deal of curiosity and a habit of getting lost in books.

Freddi lives a mostly private life, doing science by day and writing fantasy by night. They are not related to "the famous Van Goghs", but have been taken for a lost cousin often enough that they stopped correcting people.

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